

Navigating Ambition and Equity: Systemic Codependency in China's Carbon Neutrality Pathways

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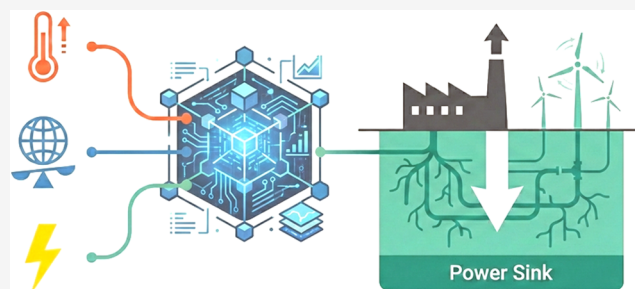
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ABSTRACT: Translating global climate targets into national decarbonization roadmaps is profoundly uncertain. To navigate this uncertainty for China, we employ a national-scale energy system model developed in the MESSAGEix framework—calibrated to China's energy balances—that uniquely combines provincial-level resolution for key sectors with a high-granularity representation of intra-annual (48 time slices) power system dynamics. Across three temperature targets (1.5, 1.6, 2.0 °C) and six allocation principles, our analysis reveals that wind and solar consistently emerge as “no-regret” pillars, CCS is essential for heavy-industry abatement, and hydrogen's sourcing shifts with budget stringency. A critical systemic codependency exists across scenarios with stringent emissions constraints: the power sector must transform into a net carbon sink to enable the decarbonization of heavy industry, creating stark path dependencies across technology choices. The 1.6 °C pathway under the Grandfathering principle presents a pragmatic alignment with China's 2060 neutrality pledge and offers a detailed blueprint for this transition. Our provincial-level analysis distinguishes high-stakes decisions from robust “no-regret” investments, offering a framework to guide China's journey to carbon neutrality.

KEYWORDS: 1.6 °C target, carbon budget, allocation principles, china, decarbonization pathways



1. INTRODUCTION

Navigating the global response to climate change is an exercise in managing profound uncertainty.¹ This uncertainty is anchored in the concept of the remaining carbon budget (RCB), which fundamentally dictates the stringency and pace of global decarbonization pathways.² The Paris Agreement's goal of limiting warming to well below 2 °C, while pursuing efforts for 1.5 °C, has framed the global ambition.³ Given the narrowing window for limiting warming to 1.5 °C with no or limited overshoot, recent literature has increasingly scrutinized pathways with low overshoot (e.g., peaking at 1.6 °C) to assess the feasibility boundaries of complying with the Paris Agreement.⁴ For any individual country, the translation of global targets into concrete action is further complicated by the allocation principles of RCB and the country's economic structure and resource endowments. Recently, ref 5 systematically quantified how the allocation principles yield vastly different emission allowances for each country, providing a foundational basis for designing national decarbonization pathways under varied temperature targets. This global context frames the central challenge for China's unprecedented journey to carbon neutrality. The question is no longer simply about finding a single path to a target, but rather: How do these layered uncertainties—spanning ambitious-yet-feasible temperature goals and contentious equity principles—affect the fundamental

robustness and structural dependencies of China's energy transition?

A substantial body of research has focused on identifying the key sectors and technologies pivotal to China's decarbonization. These studies yield valuable insights into the transition of the power sector by highlighting renewables,⁶ nuclear,⁷ BECCS,⁸ hydropower,⁹ and CCS;^{10–12} the deep decarbonization of heavy industries such as cement,^{13–15} chemical,¹⁶ and steel;^{17–19} the role of green hydrogen as a versatile energy carrier and feedstock in hard-to-abate sectors;²⁰ and the transformation of the building,²¹ heating,²² and transport^{23–25} sectors. However, these analyses mostly adopt a narrow scope, often focusing on a single sector or technology without a holistic perspective on the structural changes inherent in such a transition.²⁶ This risks underestimating the systemic impact of cross-cutting strategies, for example, demand-side electrification and decarbonization of power supply,²⁷ especially the interplay between them. More critically, this siloed approach precludes a system-wide, cost-optimal allocation of abatement efforts under a unifying carbon

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budget, which is essential for prioritizing the most economical mitigation options across all sectors. Other studies have employed integrated system models, such as Computable General Equilibrium (CGE) models^{28,29} and Integrated Assessment Models (IAMs),^{30,31} to offer a more comprehensive view. For instance, ref 32 explored China's transformations under various global effort-sharing principles for a 2 °C target, however, with limited technological details about key sectors and technologies provided. Overall, the explicit transformation roadmaps for key sectors in China, as well as the required spatiotemporal deployment of key abatement technologies, remain insufficiently explored, particularly under different climate targets. This research gap limits the availability of actionable insights for effective policy and investment planning.

This paper addresses these gaps by adopting a holistic energy system optimization framework to capture cross-sectoral dynamics and applying it to systematically explore the resilience and codependency of China's transition under uncertain global targets and allocation principles. We focus on answering two critical questions: how do these global constraints interact to shape the structural evolution of China's national emission pathways? Specifically, what are the critical systemic codependencies between sectors, and what is the required spatiotemporal diffusion of key abatement technologies to ensure a robust transition across this decision space?

To answer these questions, we employ a national-scale energy system model for China, developed within the MESSAGEix framework. The model is tailored to capture the core challenges of industrial decarbonization by incorporating the most critical sectors with process-level representations, including power and heat, iron and steel, cement, chemicals, and hydrogen production. To balance this deep sectoral granularity with computational tractability, we adopt a hybrid-resolution approach. This strategy combines provincial-level modeling for these key sectors—along with 48 representative time slices to capture intra-annual power system dynamics—with a more aggregated representation for other end-use sectors. This allows us to elucidate the dynamic, high-resolution transformation pathways for China's key sectors, thereby providing a more actionable roadmap that highlights robust investments and critical systemic trade-offs.

The rest of this paper is organized as follows. Section 2 details our modeling framework, while Section 3 explains the design of the scenario matrix used to explore key uncertainties. Section 4 presents the core findings, covering China's national emission pathways, the spatiotemporal deployment of key technologies, and a case study of the 1.6 °C pathway under the Grandfathering principle as a representative intersection of ambition and equity. The paper concludes with a discussion of the broader policy implications.

2. METHODOLOGY

This study employs a national-scale, bottom-up energy system optimization model for China, developed within the MESSAGEix framework. The framework conceptualizes the energy system—spanning production, conversion, transmission, and end-use—as a network of interconnected technologies and commodities, which can be detailed at the process level. Each technology is characterized by a set of input and output coefficients, defining the flow of different commodities (e.g., energy carriers and materials). To meet the exogenously specified final demands across all sectors, the model is formulated as a linear programming problem with perfect

foresight, minimizing the total discounted system cost by optimizing the activity levels and installed capacities of all technologies. The optimization is subject to a series of constraints, including demand-supply balances of all commodities, technology capacity expansion limits, resource availability, and policy-driven constraints like carbon budgets. refs 33,34 provide a detailed description of the model framework.

Building upon this framework, our model for China encompasses all sectors of the national energy balance, calibrated to a 2020 base year using official statistics,³⁵ with an optimization horizon spanning from 2025 to 2060 in 5-year intervals. The national energy balance statistics are aggregated into 13 distinct sectors within the model. As summarized in Table 1, the technology portfolio includes 113 backbone and 10

Table 1. Sector and Technology Coverage

sectors	backbone technology	add-on technology	
		CHP	CCS
trade	23		
afolu	5		
power and heat	26	3	4
oil refining	7		
other upstream	5		
iron and steel	8		2
chemical	9		
nonferrous metals	4		
cement	3		1
construction	5		
other downstream	4		
building	8		
transport	6		

add-on technologies. To represent the energy flows, 14 key energy carriers were selected, each accounting for at least 3% of the final energy consumption in one or more of the 13 sectors. These include coal, natural gas, diesel, gasoline, liquefied petroleum gas (LPG), heat, coke, kerosene, fuel oil, electricity, petroleum bitumen, naphtha, petroleum coke, and traditional biomass.

Finally, to balance representational granularity with computational tractability, a key feature of our model is its hybrid approach to sectoral and spatiotemporal representation, which combines high-resolution detail for sectors critical to decarbonization with a more aggregated representation for others to maintain computational tractability. The sectoral modeling is detailed as follows:

- **Power and Heat:** The power and heat sector is modeled with detailed process-level representation, encompassing a comprehensive portfolio of generation technologies, including conventional thermal power (coal, gas) and various low-carbon options (nuclear, hydro, biomass, wind, and solar). Key abatement measures like carbon capture and storage (CCS) and combined heat and power (CHP) are available as add-on technologies.¹
- **Iron and Steel:** The iron and steel sector is modeled at the process level, with seven technologies representing three main production routes: the conventional blast furnace-basic oxygen furnace (BF-BOF), the scrap-based electric arc furnace (EAF), and the hydrogen-based direct reduced iron (H2-DRI). CCS is modeled as an add-on technology for the key processes. The provincial-level disaggregation accounts for regional heterogeneity in the

feedstock prices and CCS costs. Key parameters, including techno-economic data for production processes and projections for steel product demand, are adopted from ref 37.

- **Cement:** The model represents cement production through three sequential processes: material preparation, clinker making, and cement grinding. To address process emissions, CCS is available as an abatement option for the clinker-making stage. This sector is modeled at the provincial level to reflect regional variations in the CCS costs. The underlying techno-economic parameters and future demand projections for cement are sourced from ref 37.
- **Chemicals:** The chemical industry is represented at the provincial level with a focus on ammonia and methanol, modeled with distinct process-level technologies. Other chemical products are aggregated, with the total final energy demand of the entire chemical sector calibrated to align with national energy balance statistics. The techno-economic parameters and demand projections are drawn from ref 37.
- **Other Sectors:** Other energy demand sectors are modeled in a more aggregated manner. This group includes various industries (other upstream, nonferrous metals, construction), buildings, and transport, which are represented at the national level without process-level detail; their future final energy demands are specified exogenously based on calibrations with national energy balances and policy-driven projections from the CEEGE model²⁸ (see Section 3 for details; specific demand data are provided in Section S3 of the Supporting Information). The land-use sinks of the agriculture, forestry, and land-use (AFOLU) sector are incorporated for emissions accounting purposes according to ref 38. Finally, the oil refining sector is also modeled as a single national entity, though it is represented with process-level details based on ref 37.

Apart from these sector details above, another critical input for the model is the provincial cost of CCS, which is applied across the power, steel, and cement sectors. These costs are derived from a high-resolution source-sink matching optimization model for China. This bottom-up model first compiles a comprehensive database of CO₂ sources (plant-level data for thermal power, steel, cement, and coal chemicals)^{39,40} and sinks (onshore and offshore storage at a 40 × 40 km² grid resolution).⁴¹ Using detailed cost data for the entire CCS value chain—from capture and compression to transport and storage—a Mixed-Integer Linear Programming (MILP) model is formulated to match sources with sinks at minimum cost. The solution of this model provides plant-level costs of CCS, which are then aggregated to generate the provincial-level, multisector input parameters for CCS in the model. Our baseline scenarios assume a 90% CO₂ capture rate for all CCS technologies; following recent literature advocating for higher capture rates,⁴² a sensitivity analysis examining the impact of increasing this rate to 99% is presented in Section S4 of the Supporting Information.

Hydrogen production is modeled at the provincial level, encompassing pathways based on coal, natural gas, and water electrolysis. The techno-economic parameters for these production routes are drawn from ref 43. A key model assumption is that hydrogen is consumed locally within the province of its production. In contrast to the “copper-plate”

assumption for electricity, interprovincial transport of hydrogen is precluded—a simplification justified by its high transport costs and designed to manage the scale of the optimization problem. Hydrogen serves multiple end-uses: as a reducing agent for the steel industry, a feedstock for chemical synthesis, and an energy carrier for industrial heat, where it competes directly with coal and natural gas. From an emissions accounting perspective, hydrogen is not treated as an independent sector in the model. Instead, emissions from hydrogen production are attributed to the sectors where hydrogen is consumed: when hydrogen is produced from coal or natural gas, the associated CO₂ emissions are counted in the corresponding end-use sectors (e.g., steel, chemicals, industrial heat).

In summary, the modeling philosophy focuses on representing core decarbonization sectors with high granularity, while other resource inputs or sectors’ final energy demands are simplified using “import” or “tube” technologies. The combination of provincial disaggregation for key sectors and the intra-annual temporal resolution (48 time slices) results in a large-scale optimization problem with 1,845,649 variables for activity alone. A single scenario requires 3–4 h to solve on a MacBook M4 Pro machine, with a peak memory usage of approximately 25 GB. Notably, this version of the model deliberately excludes a suite of 107 energy efficiency add-on technologies for the steel, cement, and oil refining sectors. While a more complex configuration including these options saw its memory footprint reach approximately 280 GB, the optimization results showed minimal deployment of these efficiency measures. This is because they primarily reduce electricity consumption, which has a diminishing impact on carbon abatement in a future power system dominated by renewables. Therefore, to maintain computational tractability and focus on the most critical decarbonization levers, these energy efficiency options were omitted from the final analysis. It is important to clarify that this exclusion applies only to specific add-on retrofit measures within the process-level sectors (e.g., waste heat recovery). It does not preclude systemic efficiency gains arising from structural technology shifts (e.g., from blast furnaces to electric arc furnaces). Furthermore, for sectors modeled with exogenous demands (buildings, transport), continuous energy efficiency improvements are implicitly embedded in their projected demand trajectories, particularly through the efficiency advantages of electrification.

3. SCENARIO DESIGN

To systematically navigate the deep uncertainties in China’s decarbonization pathways and answer the research questions posed above, we construct a comprehensive scenario matrix of 54 plausible futures. These scenarios are designed to map the decision space defined by three key dimensions of uncertainty: the stringency of the global climate target, the principle for international carbon budget allocation, and the pace of domestic end-use electrification. This multidimensional framework allows us to identify structural evolutions and systemic codependencies that result from the interactions of these uncertain drivers.

The first dimension considers three long-term global temperature targets. Alongside the Paris Agreement’s established 1.5 and 2.0 °C goals, we include a 1.6 °C target as an intermediate benchmark. The 1.5 °C target imposes stringent budgets and leaves minimal room for feasible transition pathways in large developing economies with substantial industrial inertia, while the 2.0 °C target, though more accommodating, risks delaying critical structural transforma-

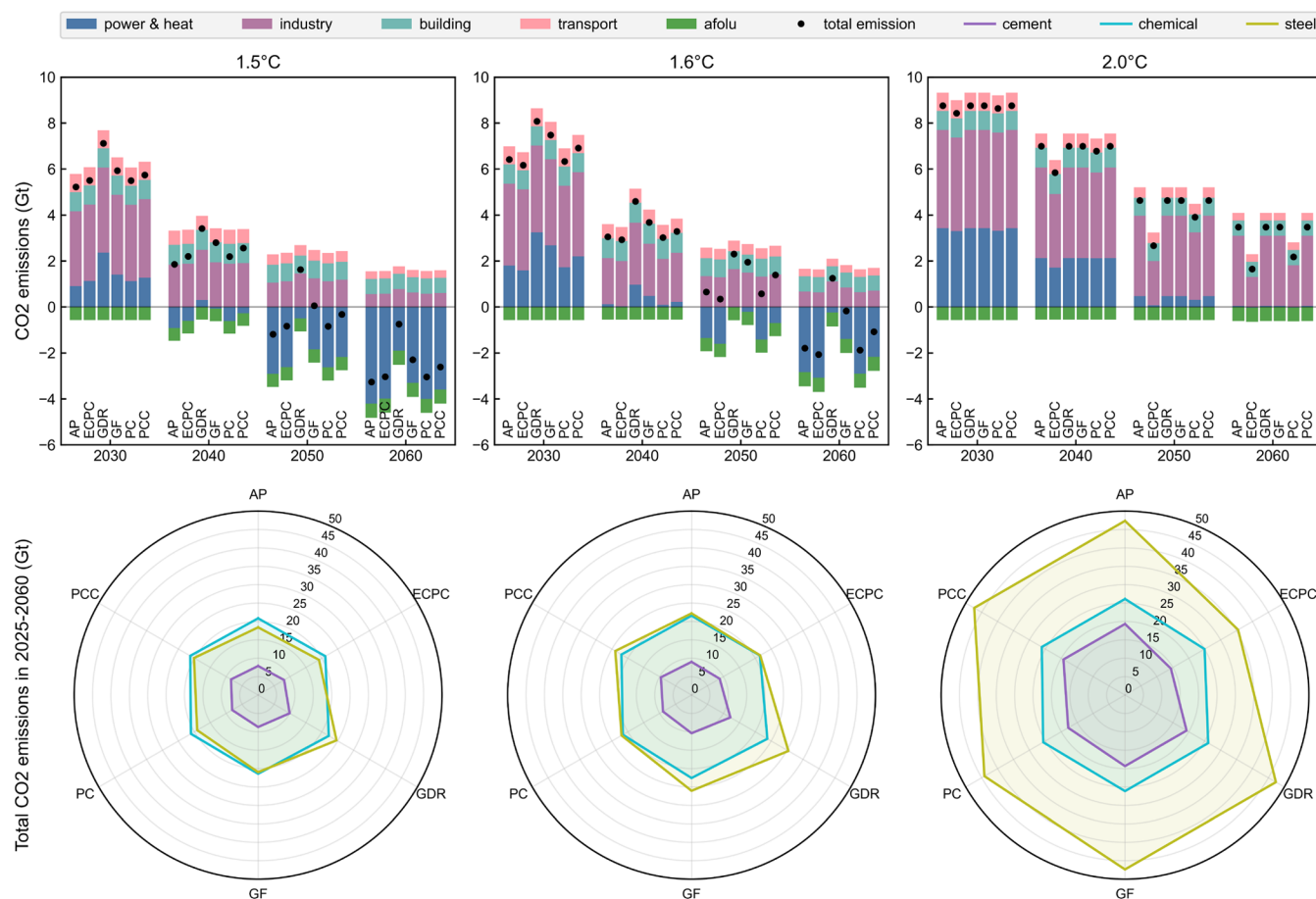


Figure 1. Sectoral and industrial CO₂ emissions under alternative temperature targets and allocation principles. Top row: Stacked bars show annual CO₂ emissions by sector (power and heat, industry, buildings, transport, afolu) under various target temperatures and allocation principles in 2030, 2040, 2050, and 2060; black dots denote total emissions, and negative values indicate net removals. Bottom row: Radar charts report cumulative CO₂ emissions in 2025–2060 from heavy industry (cement, chemical, steel).

tions needed to avoid carbon lock-in. The 1.6 °C target thus occupies a pragmatic middle ground that maintains high climate ambition while providing a more realistic space for exploring national transition dynamics under different equity considerations. Specifically, the carbon budgets for all three targets correspond to a 50% likelihood of limiting peak warming to these levels. It is important to note that our 1.6 °C scenario treats this budget as a hard constraint for stabilization, distinct from “overshoot” pathways (e.g., as discussed in ref 4) that might exceed this temperature temporarily and require deeper subsequent negative emissions. The second dimension addresses the contentious issue of equity in global climate action by applying six distinct principles to allocate China’s national share of the remaining global carbon budget, following the framework of ref 5. These principles encapsulate a wide spectrum of ethical considerations, ranging from historical responsibility to per capita equality, including: Grandfathering (GF), which preserves current emission fractions; Per Capita (PC), allocating emissions based on population; Per Capita Convergence (PCC), transitioning from GF to PC over time; Ability to Pay (AP), allocating emissions inversely to GDP per capita; Equal Cumulative Per Capita (ECPC), which reconciles historical emissions before converging to a PC basis; and Greenhouse Development Rights (GDR), based on a Responsibility-Capability Index. The combination of these

temperature targets and allocation principles defines 18 distinct carbon budget trajectories for China from 2025 to 2060.

The third dimension explores uncertainty in domestic demand-side transformation through three levels of future electrification for sectors where technological change is not endogenously modeled (e.g., buildings and transport). These levels represent different policy and technology adoption narratives: a **high** level assumes rapid technology diffusion (e.g., of electric vehicles and heat pumps) driven by strong policy support; a **medium** level represents a continuation of current trends; and a **low** level considers scenarios with slower adoption due to potential barriers. These trajectories are not abstract assumptions but are calibrated against detailed projections from the CEEGE model,²⁸ providing internally consistent pathways of final energy demand by carrier as exogenous inputs to our model. In this way, China-specific policy drivers are captured through a hybrid approach: for sectors modeled endogenously at the process level, the binding carbon budget constraint functions as an implicit carbon price, simulating the effect of carbon pricing mechanisms (e.g., ETS) and administrative mandates that force the adoption of low-carbon technologies; for sectors with exogenously specified demands, policy impacts such as electrification promotion and energy efficiency improvements are embedded in the demand trajectories inherited from the CGE model.

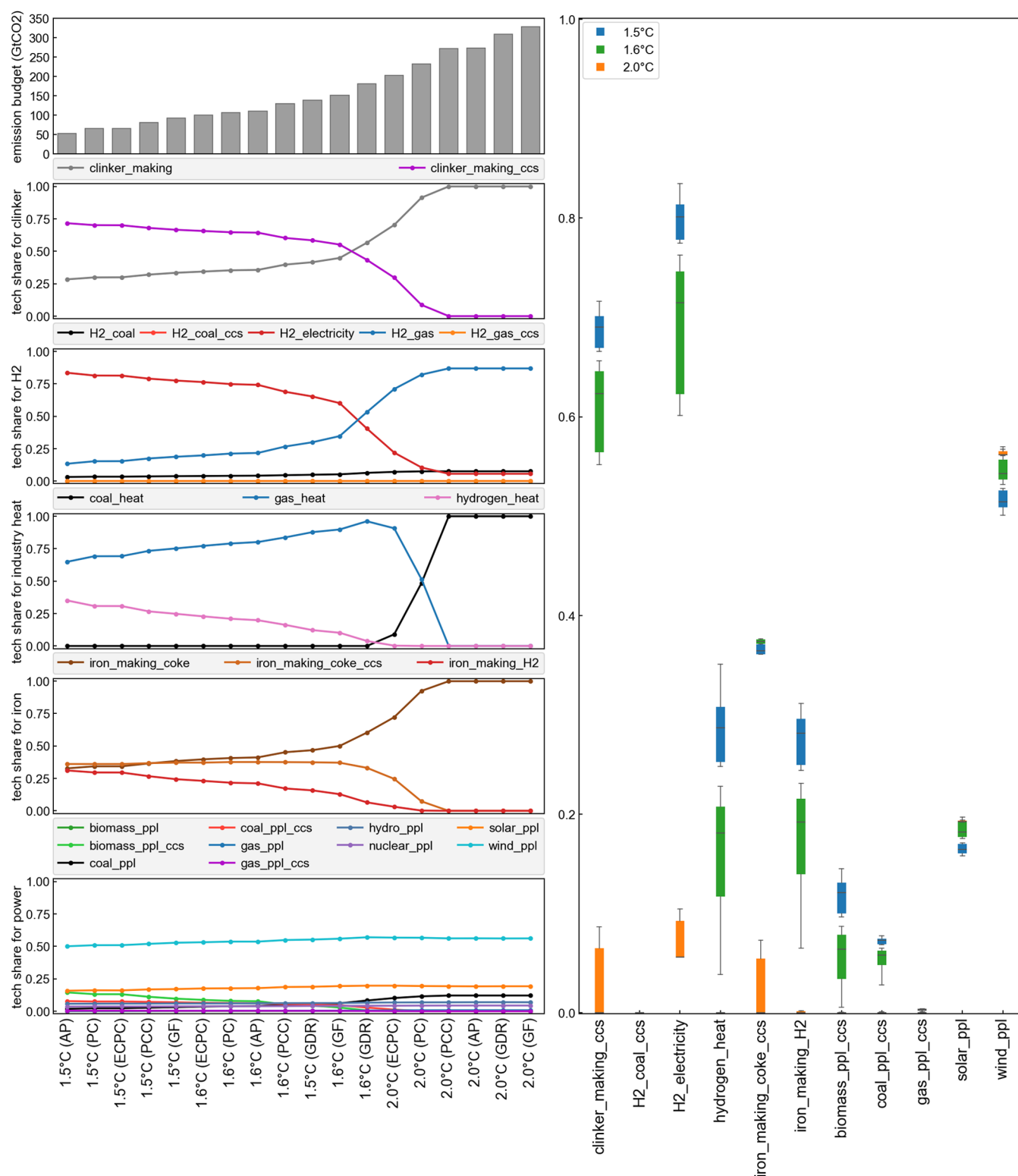


Figure 2. Technology portfolios under alternative temperature targets and allocation principles. Left: The top panel shows China's cumulative CO₂ budget for 2025–2060 for each scenario, which combines a temperature target and an allocation principle. The subsequent panels report the cumulative technology shares for (i) clinker production, (ii) hydrogen production, (iii) industrial heat, (iv) ironmaking, and (v) power generation. Right: Boxplots summarize the variation in cumulative shares arising from the different allocation principles for each temperature target (blue: 1.5 °C, green: 1.6 °C, orange: 2.0 °C).

The resulting $3 \times 6 \times 3$ matrix of 54 scenarios provides a robust framework for exploring the solution space. For clarity of visualization in Figure 1, 2, 3, 4 and 5, results are averaged across the three electrification levels to isolate the impact of carbon

budget uncertainty. Sensitivity analyses with respect to the electrification level are provided in Sections S1 and S2 of the Supporting Information. Subsequently, the impact of varying electrification levels is explicitly analyzed in Figure 6, 7 and 8 for

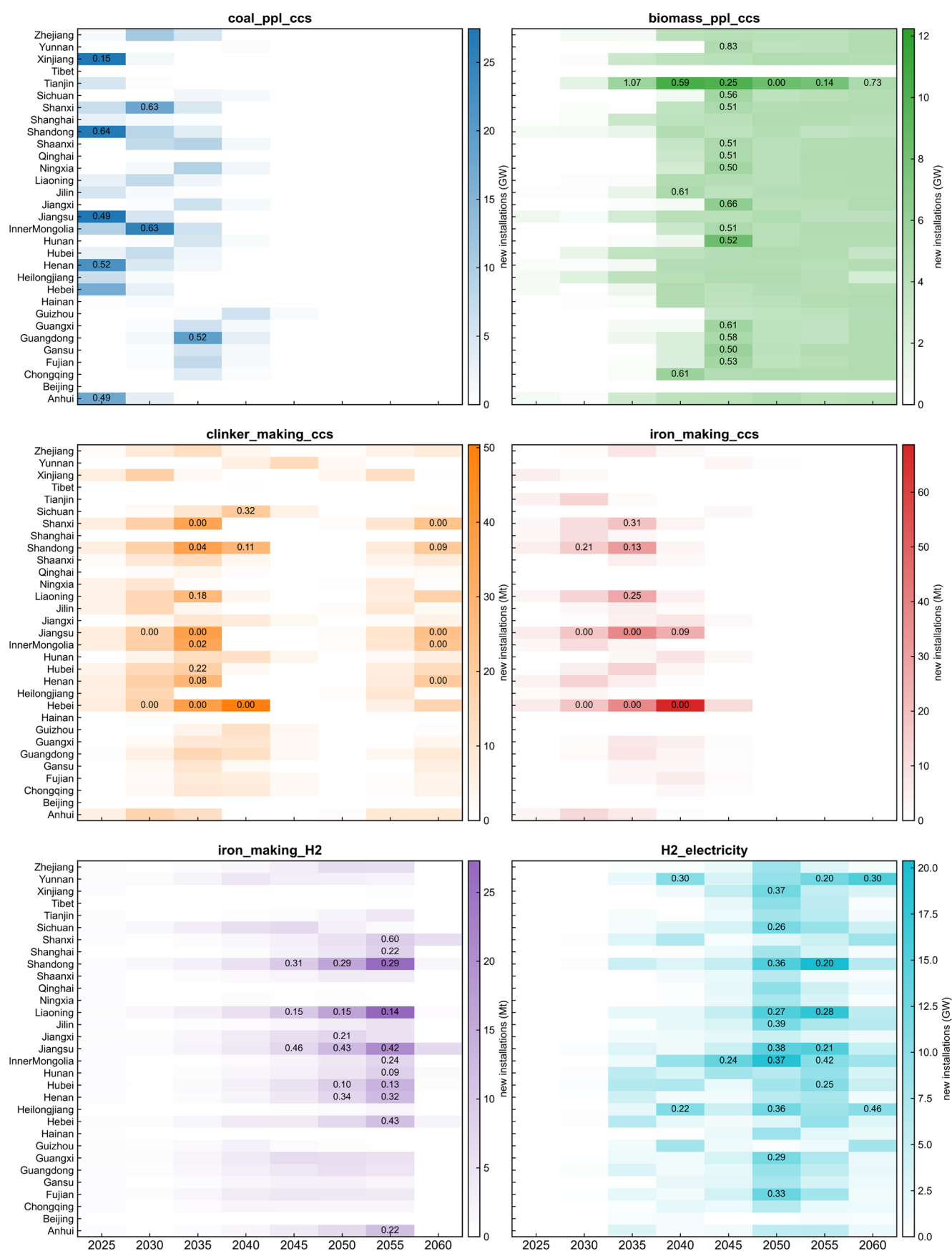


Figure 3. Spatiotemporal deployment of key technologies under the 1.5 °C target. The six panels correspond to coal power with CCS, biomass power with CCS, clinker with CCS, ironmaking with CCS, H₂-DRI ironmaking, and electrolytic H₂. The heatmaps show the mean annual new capacity installations by province (y-axis) and year (x-axis, 2025–2060). Color intensity denotes the scale of deployment (units in the colorbar to the right of

Figure 3. continued

each panel; GW or Mt). Cells in the top decile of positive values are annotated with the coefficient of variation (std/mean) across allocation principles, indicating planning uncertainty.

the 1.6 °C pathway under the Grandfathering principle, offering a deeper dive into the interplay between supply- and demand-side transformations.

4. RESULTS AND DISCUSSION

4.1. China's Emissions Pathways across Global Warming Scenarios

Following ref 5, we calculated the sectoral decarbonization pathways for China under three temperature targets (1.5 °C, 1.6 °C, and 2.0 °C) and six allocation principles. The carbon budget under different scenarios are downloaded from the Carbon-BudgetExplorer² for the years from 2025 to 2060.

As shown in Figure 1, China's decarbonization pathways are shaped by the stringency of the global temperature target and the choice of allocation principle. A clear pattern emerges across all scenarios: looser temperature targets systematically correspond to larger annual emission allowances and higher cumulative budgets for heavy industry (top and bottom panels, respectively). The 1.5 °C pathways, in particular, demand exceptionally deep emission cuts, mandating that the entire economy achieves carbon neutrality no later than 2050. Within a given temperature target, the allocation principle acts as a critical variable, leading to vastly different emission trajectories and sectoral responsibilities. Despite this variability, a common strategy is the pivotal role of the power and heat sector, which must deploy negative-emission technologies at scale in later decades to offset residual emissions from hard-to-abate sectors. This underscores the increasing reliance on large-scale CO₂ removal, especially under the more ambitious 1.5 and 1.6 °C targets. Notably, among these diverse trajectories, the 1.6 °C pathway under the GF principle results in slightly negative emissions by 2060, closely matching China's carbon neutrality pledge.

A deeper analysis of Figure 1 reveals a crucial codependency between the power sector and heavy industry that defines China's transition. The pathways show the power sector undergoing a radical transformation, shifting from the largest emissions source to the primary engine of carbon removal via technologies like BECCS, particularly after 2040. This pivot is the critical enabler that creates the necessary carbon space for heavy industry to manage its stubborn residual emissions. The radar charts quantify this dependency: the constrained industrial emission budgets under the 1.5 and 1.6 °C targets are only feasible because of the massive carbon sink created by the power sector. This framing shifts the challenge from a simple sum of sectoral cuts to a systemic balancing act, necessitating a specific portfolio of technologies to manage this intersectoral carbon offsetting.

4.2. Key Technology Portfolios and Spatiotemporal Deployment

The technological backbone for the ambitious emission pathways outlined above must be formed by a strategic portfolio of key technologies.

As presented in Figure 2, tighter budgets unequivocally accelerate the adoption of a synergistic suite of low-carbon technologies across key industrial and energy sectors. The role and sourcing of hydrogen, for instance, are highly sensitive to the

available budget. Under the most stringent 1.5 °C pathways, green hydrogen from electrolysis becomes the dominant production route, accounting for up to 83% of total supply and serving as an essential decarbonization lever for both industrial heat and ironmaking (via H₂-DRI).

Under the looser 2.0 °C scenarios, however, this trend reverses sharply: the share of electrolysis plummets to as low as 6%, while natural gas-based hydrogen expands to supply the nearly 87% remainder. This upstream energy shift is mirrored in the steel industry, where hydrogen-based ironmaking (H₂-DRI) is phased out in favor of traditional blast furnaces. This stark trade-off highlights a critical path dependency: near-term investments favoring large-scale electrolysis and renewable capacity under stringent targets lock in a green hydrogen economy, whereas channeling capital into natural gas reforming under looser budgets prolongs the transition away from fossil fuels.

Carbon capture and storage emerges as another key enabling technology, but its role is nuanced and sector-dependent. For cement production, CCS is a robustly essential strategy under 1.5 and 1.6 °C targets, where it is applied to over 70% of clinker production. In ironmaking, CCS applied to traditional blast furnaces is a significant transitional technology, maintaining a share of around 36% in these scenarios, but it is entirely phased out in 2.0 °C pathways. Its role in power generation is more limited, primarily serving to extend the lifetime of some coal power plants in pathways with larger carbon allowances. The power sector forms the backbone of the transition, with wind and solar power as the undisputed pillars, collectively constituting between 66% and 75% of total generation across all scenarios. The primary distinction between pathways lies in the composition of the dispatchable, low-carbon fleet. Tighter budgets necessitate a heavy reliance on BECCS, which provides up to 14.5% of electricity generation in 1.5 °C scenarios, making it the third-largest contributor. This critical role in providing dispatchable, negative-emission power partially crowds out wind and solar technologies. As the budget loosens, the role of BECCS diminishes, disappearing entirely in 2.0 °C pathways, while hydropower and nuclear power provide stable contributions to the grid.

Beyond these broad trends, the boxplots in Figure 2 provide a granular map of the investment uncertainty introduced by allocation principles. As shown, the deployment shares for renewable power—particularly solar and wind—exhibit narrow ranges across all scenarios. This marks them as the unambiguous “no-regret” bedrock of the transition.

In stark contrast, technologies like BECCS in the power sector and green hydrogen production face profound uncertainty. For instance, under the 1.6 °C target, the required share of BECCS can vary by over 10-fold (from 0.5% to 8.7%) depending on the allocation principle. Furthermore, another crucial insight is that loosening the climate target can increase planning uncertainty. For cement with CCS, for instance, the planning uncertainty increases as the climate target loosens from 1.5 to 2.0 °C, because the tight budget under the 1.5 °C target effectively locks in large-scale CCS as a necessity regardless of the allocation principle, thus reducing uncertainty. In contrast, the looser 2.0 °C budget creates more strategic leeway, making the decision to

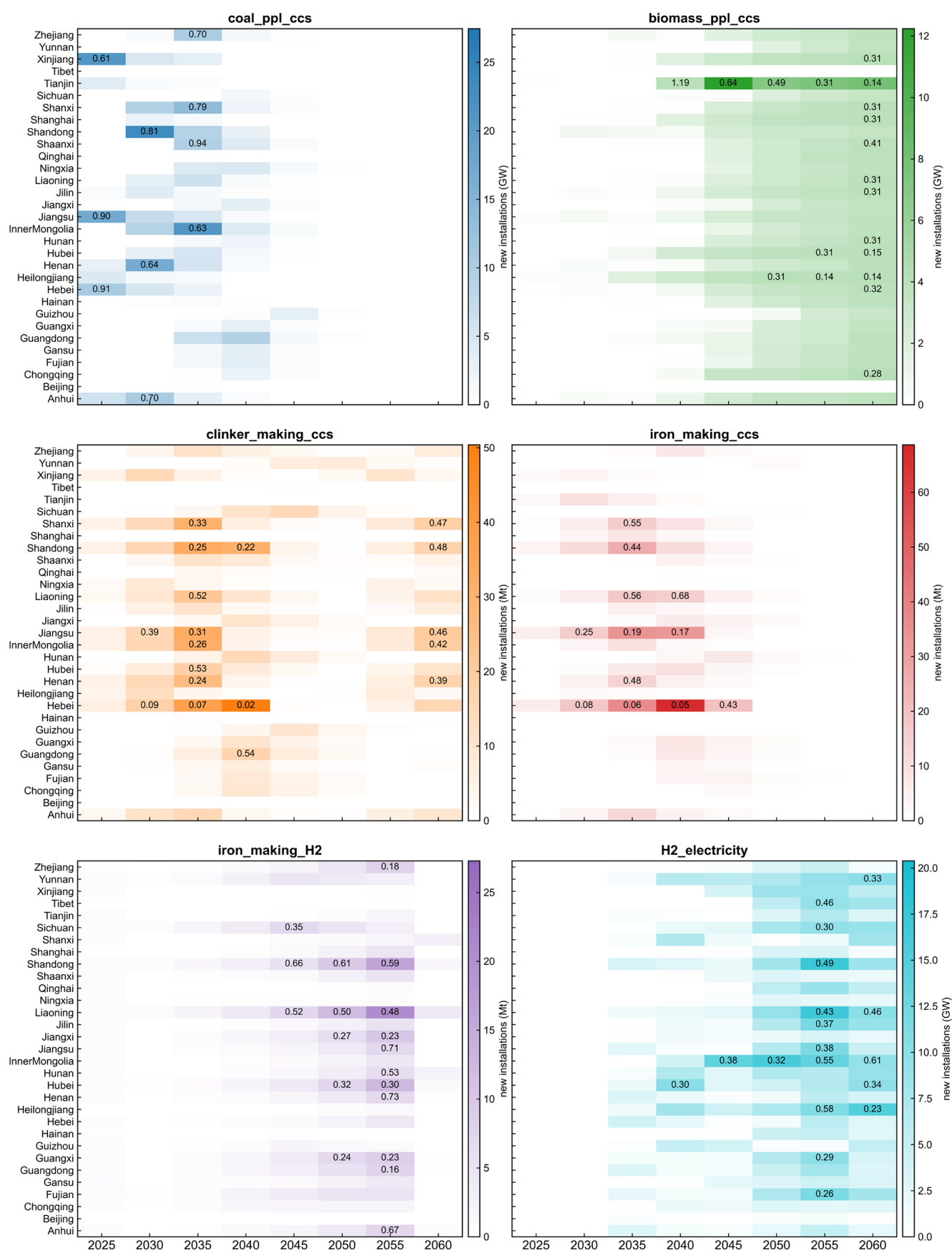


Figure 4. Spatiotemporal deployment of key technologies under the 1.6 °C target. Panel and annotation conventions follow Figure 3.

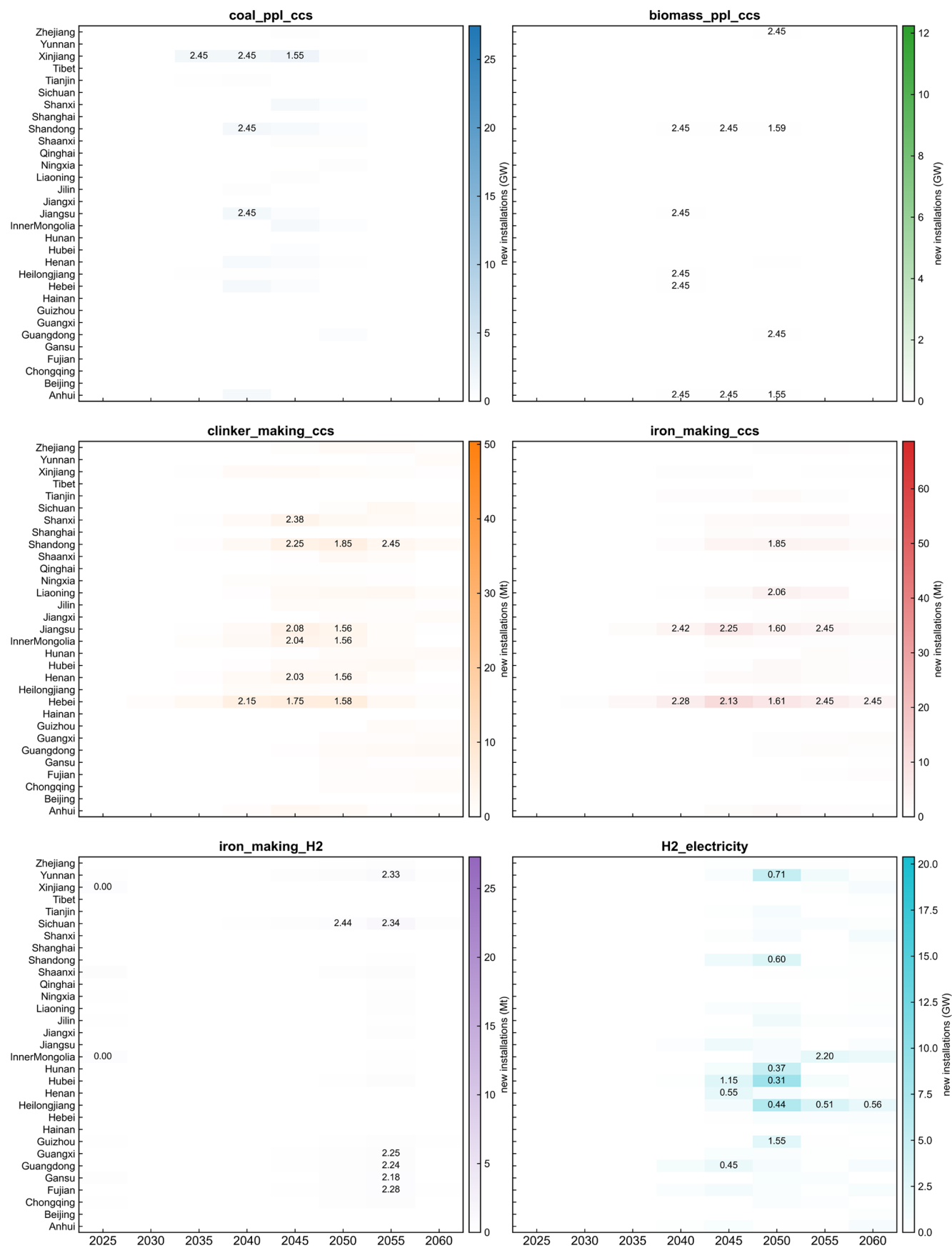


Figure 5. Spatiotemporal deployment of key technologies under the 2.0 °C target. Panel and annotation conventions follow Figure 3.

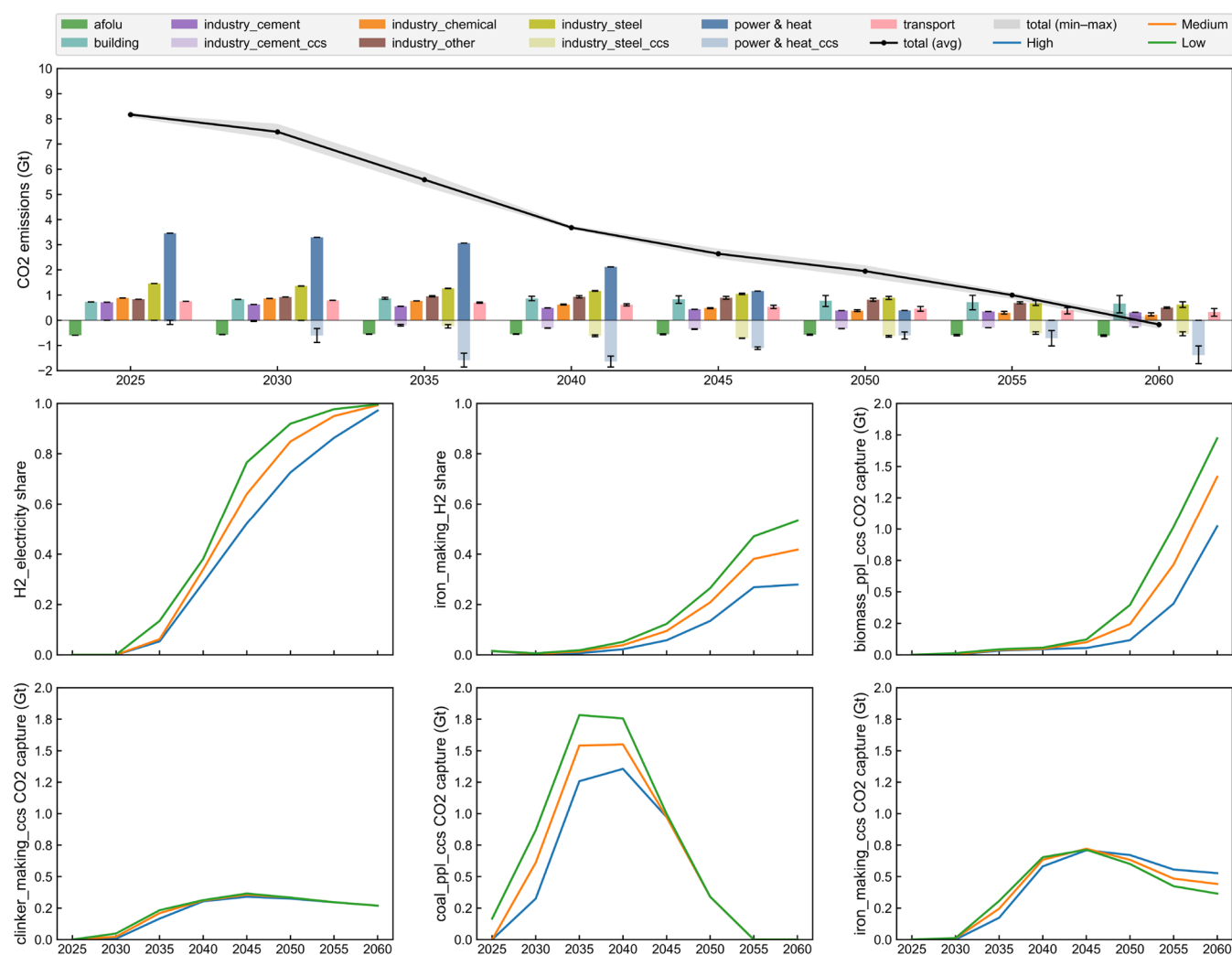


Figure 6. Emissions pathway and technology responses for China under the 1.6 °C target with the GF allocation principle. Top: Stacked bars show annual CO₂ emissions by sector from 2025 to 2060. The black line denotes total emissions averaged across electrification scenarios, with the gray band indicating the range; negative bars indicate net removals. Bottom: Trajectories of key technology variables by electrification level (green high, orange medium, blue low): share of electrolytic H₂ and H₂-DRI, and annual CO₂ captured by biomass power CCS, coal power CCS, clinker CCS, and ironmaking CCS.

deploy CCS highly contingent on the chosen principle, thereby increasing the planning uncertainty.

Finally, having identified the strategic technology portfolios, Figures 3–5 map out their spatiotemporal deployment. A clear, overarching trend is that stricter temperature targets necessitate an earlier, larger-scale, and more widespread deployment across the country. Crucially, the annotated cells in these deployment hotspots also reveal a considerable planning uncertainty, quantified by the coefficient of variation (std/mean). This metric offers a clear guide for investment: intensely colored cells with low variation represent robust, high-priority opportunities, whereas those with high variation highlight high-risk investments whose required build-out pace is highly sensitive to the chosen allocation principle. This planning uncertainty is notably reduced under stricter temperature targets, as the imperative for rapid decarbonization narrows the range of viable options. Furthermore, the strong colocation of electrolytic hydrogen production and H₂-DRI ironmaking suggests the potential formation of industrial hubs where green hydrogen is produced locally to decarbonize adjacent industrial facilities. It is important to note that this observed colocation is reinforced

by our assumption excluding interprovincial hydrogen transport. Realizing a national hydrogen network involves complex trade-offs between accessing lower-cost western resources and incurring substantial transport infrastructure costs. Moreover, regional renewable potentials impose physical constraints on local production. Consequently, the net impact of these opposing factors on the marginal cost of supply—and thus the competitiveness of hydrogen-based ironmaking—remains uncertain without detailed modeling.

4.3. Analyzing the 1.6 °C-Grandfathering Scenario: A Representative Case of Ambition-Equity Intersection

As shown in Figure 1, China's CO₂ emission in 2060 under the 1.6 °C target with the GF allocation principle is −0.169 Gt, close to 0, which closely matches China's 2060 neutrality pledge. While other pathways could also theoretically achieve neutrality, we select this scenario as a representative case study as it harmonizes the global imperatives of ambition and equity with China's domestic strategic timeline.

Figure 6 details the 1.6 °C-GF pathway and shows the impact of end-use electrification levels. We vary the electrification level because several demand sectors are not modeled at the process

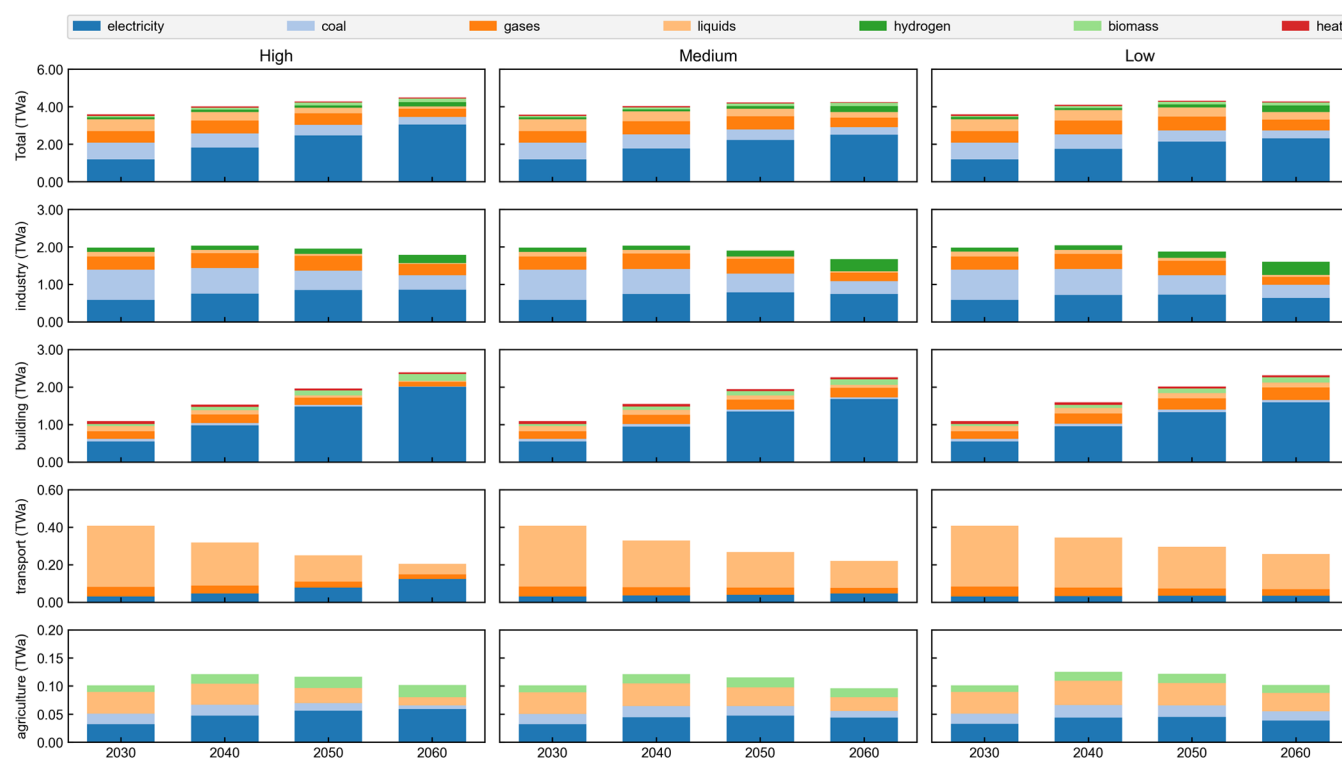


Figure 7. Final energy demand by sector and carrier under the 1.6 °C target with the GF allocation principle. Columns denote electrification levels (High, Medium, Low); rows report Total, Industry, Buildings, Transport, and Agriculture. Stacked bars show demand in 2030, 2040, 2050, and 2060 by carrier.

level. Their final energy trajectories are specified exogenously based on the results from a CGE model, CEEGE²⁸ (see Section 3 for details). As shown, across electrification levels, residual emissions from industry and transport must be offset by removals from the power and AFOLU sectors. By 2060, BECCS in the power sector supplies large-scale removals (order of 1–2 Gt CO₂/year across levels). Electrification rebalances the technology mix: higher electrification accelerates and lifts the trajectories of electrolytic hydrogen and H₂-DRI, and raises cumulative removals (BECCS). Conversely, lower electrification tightens the residual carbon space within the process-level sectors and induces a stronger pivot toward H₂/H₂-DRI, while BF-CCS in steel declines as it is displaced. Overall, differences across pathways are expressed mainly as a redistribution of abatement effort between power and industry. A significant finding is the transitional role of CCS on coal-fired power plants: its deployment peaks around 2035–2040 and then declines as the coal fleet is phased out. By contrast, CCS in industrial processes and BECCS are sustained as long-term strategic options; clinker CCS is indispensable, while ironmaking-CCS gradually gives way to H₂-DRI as electrolysis scales.

The deep decarbonization of the energy system is contingent upon a profound transformation of the final energy demand across all major sectors (Figure 7). A clear and universal trend is the rise of electrification: electricity steadily replaces fossil fuels to become the dominant energy carrier in industry, buildings, and agriculture, with the building sector approaching near-total electrification by 2060. The industrial sector, as the largest energy consumer, acts as the anchor of this transition, exhibiting a complex fuel-switching dynamic, where coal is progressively phased out in favor of electricity, natural gas, hydrogen, and biomass. In stark contrast, the transport sector remains heavily reliant on liquid fuels, highlighting the persistent challenge of

decarbonizing aviation, shipping, and heavy-duty freight. The level of electrification significantly influences the pace and depth of this transition. A high-electrification scenario accelerates the phase-out of coal, drives higher electricity consumption across the board, and enables an earlier and larger-scale deployment of hydrogen, particularly in industry. Notably, despite continued economic growth, total final energy consumption is projected to plateau and even decline after 2040, underscoring the critical role of electrification-induced efficiency gains and exogenous demand projections reflecting continuous efficiency improvements in decoupling economic activity from energy demand.

The evolution of the power system, as the linchpin of the entire energy transition, is detailed in Figure 8. A complete structural shift is projected, with wind and solar power emerging as the undisputed pillars of the future electricity supply. Their contribution grows rapidly from 2025, constituting the vast majority of the generation by 2060. Concurrently, the role of coal power diminishes over time, marking an orderly phase-out that culminates in its near-total exit from the generation mix by midcentury. The bottom panels, illustrating the intra-annual dispatch in 2060, depict the operational reality of a system dominated by variable renewables. The distinct daily peaks of solar generation, particularly in spring and summer, create a pronounced “duck curve” that necessitates a portfolio of flexible resources for system balancing. This portfolio includes hydro-power, nuclear, and biomass, which provide stable baseload power, thereby ensuring reliability during periods of low wind and solar output. The level of electrification directly determines the scale of the power system; a high-electrification scenario requires a significantly larger generation capacity to meet the increased demand from other sectors. Ultimately, the figure showcases the operational dynamics of a next-generation power system, one that relies on a complementary mix of variable

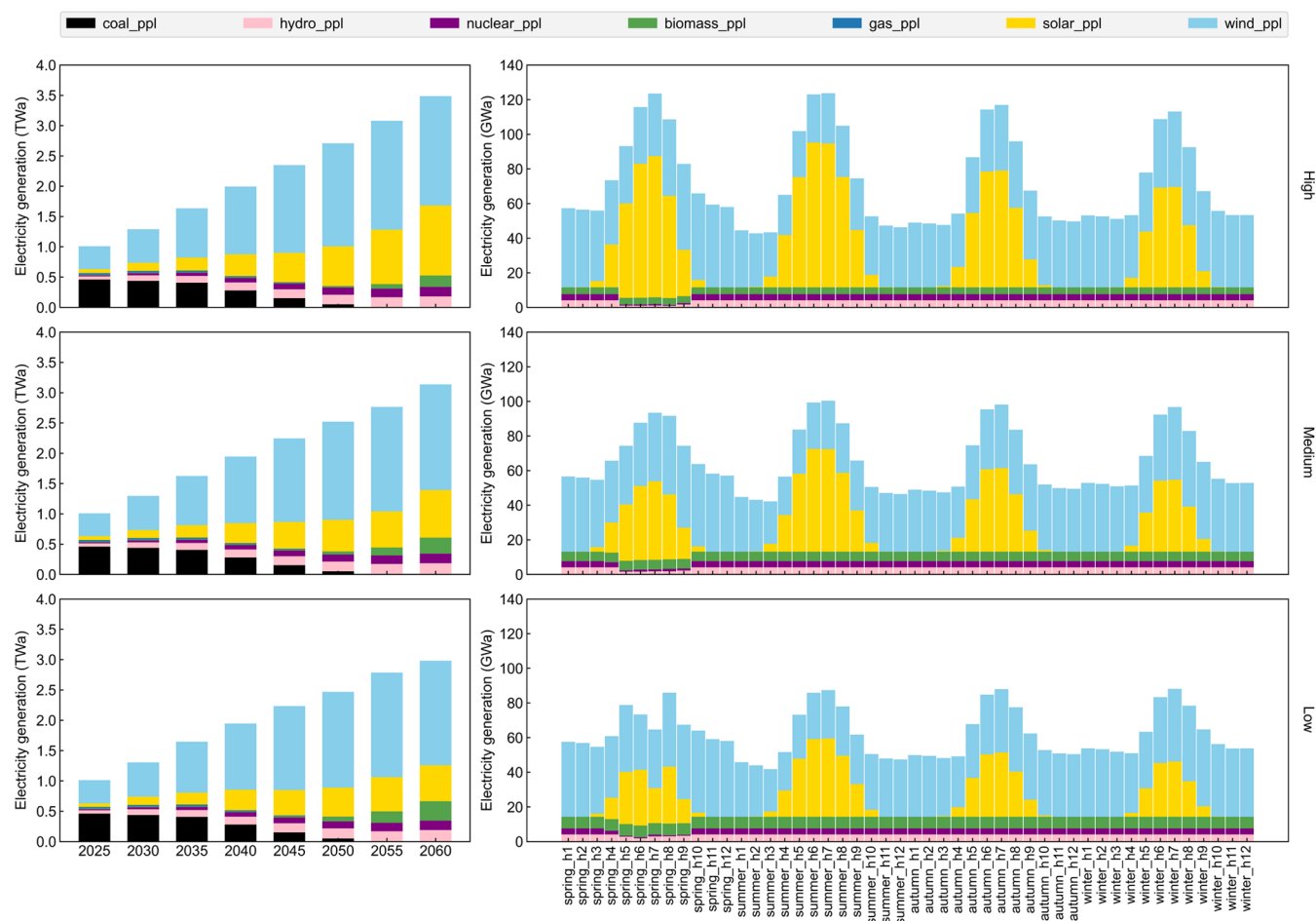


Figure 8. Power system evolution under the 1.6 °C target with the GF allocation principle. Rows denote electrification levels (High, Medium, Low). Left column: annual electricity generation by technology (coal, gas, biomass, hydro, nuclear, solar, wind) from 2025 to 2060 (stacked; units: TWh). Right column: intra-annual generation mix in 2060 for four representative days (spring, summer, autumn, winter) at 2 h resolution (units: GWh), illustrating diurnal/seasonal variability and the complementary roles of variable renewables and flexible resources.

renewables and dispatchable, low-carbon resources to manage seasonal and diurnal variability.

4.4. Discussion

By systematically quantifying China's decarbonization pathways under the dual uncertainties of global climate targets and international equity principles, this study provides critical insights for navigating national strategy with long-term climate goals. Our analysis identifies that wind and solar power are the “no-regret” pillars of the future energy system, while carbon capture is essential for deep decarbonization of heavy industry. Under the ambitious 1.5 and 1.6 °C targets, this transition further requires the power sector to transform from the largest emitter into a net carbon sink. The 1.6 °C pathway under the Grandfathering (GF) principle, which closely matches China's 2060 neutrality pledge, reveals a nuanced technology strategy. While not the only feasible route, this scenario serves as a critical reference point by crystallizing the intersection of three key dimensions: global climate ambition, international equity, and national strategic timelines. It is characterized by a complete phase-out of unabated coal power, with CCS on the remaining fleet serving only a transitional role that peaks around 2040. In contrast, CCS on industrial processes is sustained as a long-term solution. Ultimately, neutrality is achieved by deploying BECCS at scale to deliver 1.5 Gt of negative emissions by 2060, compensating for residual emissions from hard-to-abate sectors.

These findings underpin three core insights. First, China's transition is not a simple summation of sectoral efforts but a systemic balancing act, defined by a crucial codependency between the power sector and heavy industry. The power sector must undergo a radical transformation—from the largest emitter to the primary engine of carbon removal—to create the necessary carbon space for hard-to-abate industries. Second, this intersectoral offsetting is technologically specific and highly sensitive to budget constraints, revealing critical path dependencies. Stricter budgets favor a pivot to green hydrogen and extensive CCS, while looser budgets allow for continued reliance on fossil-based routes, highlighting the immense long-term consequences of near-term investment choices. Third, planning uncertainty cascades from the national to the provincial level, but it is not uniformly distributed. Our spatiotemporal analysis pinpoints regional hotspots for technology deployment and, crucially, offers a quantitative metric (the coefficient of variation) to distinguish robust, “no-regret” investments from high-risk ventures that are contingent on specific allocation principles. As a final consideration, while our deterministic optimization framework provides a clear view of cost-optimal pathways, it is important to acknowledge that real-world investment decisions will also need to navigate uncertainties not captured in the model, such as future technological breakthroughs and market dynamics.

Crucially, these modeling outcomes translate into distinct policy imperatives depending on the interplay between global ambition and equity. Under a looser 2.0 °C target, the primary driver of uncertainty is the choice of allocation principle. Principles recognizing development rights (e.g., GDR) or historical inertia (e.g., Grandfathering) grant China larger carbon budgets, allowing for an “orderly transition” focused on cost minimization through natural asset turnover and retrofitting. In contrast, population-based principles (e.g., Per Capita) would still impose significant constraints. This divergence persists into the 1.6 °C scenarios, where GDR still provides a strategic buffer that allows for a smoother adjustment compared to the “wartime” contraction necessitated by population-based principles. However, under the strictest 1.5 °C target, the policy space narrows dramatically. The imperative for rapid decarbonization overrides the nuances of allocation principles, necessitating immediate, aggressive administrative mandates for the early retirement of fossil assets and massive subsidies for precommercial technologies like H₂-DRI and BECCS across all equity scenarios. Given this geopolitical uncertainty, a “two-track” strategy emerges as the most robust approach: prioritizing the “no-regret” massive expansion of renewables and grid flexibility—which is universally required—while simultaneously hedging risks by establishing pilot zones to test deep-decarbonization technologies (e.g., green hydrogen hubs) that may be urgently needed if global climate regimes shift toward stricter targets or equity interpretations.

Finally, this study focuses exclusively on energy-related CO₂ emissions, reflecting the scope of our model. However, we acknowledge that non-CO₂ greenhouse gases (e.g., methane, N₂O) and land-use change emissions are critical for full climate neutrality. Incorporating them would effectively tighten the carbon budget available for the energy sector. Experiences from our stricter scenarios (e.g., 1.5 °C) suggest that this would further accelerate the deployment of deep-decarbonization technologies, such as industrial electrification and hydrogen, alongside the expansion of carbon removal capacities. Future work will aim to couple this model with land-use modules to fully capture the broader systemic codependencies across energy and land systems, particularly the potential crowding-out effects of large-scale bioenergy deployment and associated land-use change emissions. Another limitation lies in the fact that for sectors modeled with exogenous demands (e.g., transport), the technology mix—such as the penetration of hydrogen fuel cell vehicles—is inherited from the underlying macroeconomic scenarios rather than being endogenously optimized. This highlights the need for fully integrated, multisector system optimization in future work to better capture these cross-sectoral technology competitions.

■ ASSOCIATED CONTENT

Data Availability Statement

The code of the model used in this study is open-source at https://github.com/iiasa/message_ix. The data can be shared on request.

SI Supporting Information

The Supporting Information is available free of charge at <https://pubs.acs.org/doi/10.1021/acs.est.5c15542>.

Contains sensitivity analyses on electrification levels for national emissions pathways (Section S1) and key technology portfolios with spatiotemporal deployment (Section S2), exogenous demand projections used in the

model (Section S3), and a sensitivity analysis on CCS capture rates comparing 90% versus 99% scenarios (Section S4) (ZIP)

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Author Contributions

S.Y. and H.D. conceived the research. S.Y. developed the model, based on the industry sectors data collected by F.Z., CGE scenarios created by L.Y., power sector data collected by X.Y., and CCS cost scenarios created by Z.L. and N.W., S.Y., and H.D. conducted the analysis, interpreted the results, and wrote the paper with inputs from the other authors.

Notes

The authors declare no competing financial interest.

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■ ADDITIONAL NOTES

¹This study models BECCS as dedicated biomass power plants with CCS rather than coal-biomass cofiring retrofits (CBECCS). The technology is omitted for two reasons. First, a recent study³⁶ suggests that CBECCS adoption remains minimal (approximately 0.1%) even in plant-level optimizations where renewables are not competing endogenously. In our system-wide optimization, coal-based technologies face even stronger economic headwinds against wind and solar. Second, at

typical cofiring ratios, CBECCS cannot deliver the net-negative emissions required to offset residual emissions from hard-to-abate sectors—the captured CO₂ is predominantly from coal combustion.

²Dekker, M.M. (2023) The Carbon Budget Explorer. [online]. Available: <https://www.carbonbudgetexplorer.eu>.

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